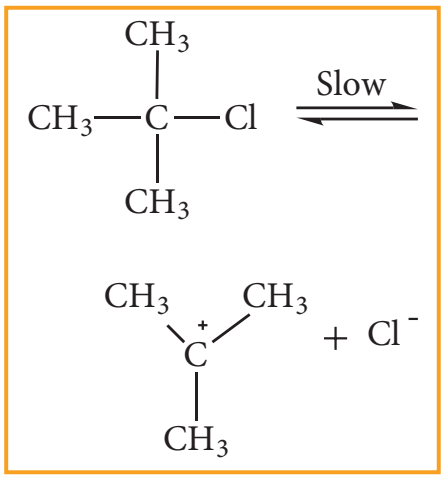
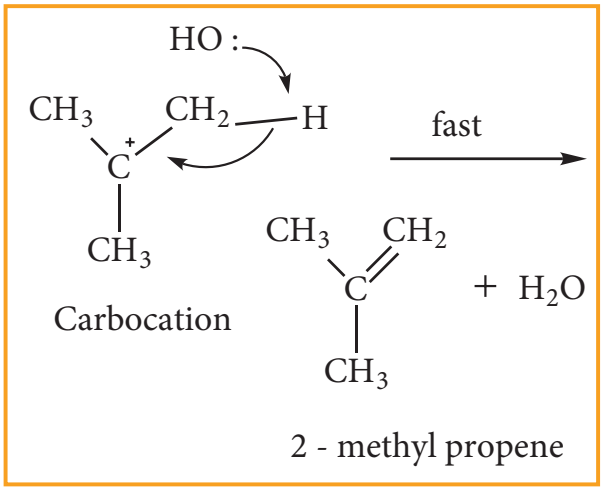
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Step - 2 Elimination of a proton from the β - carbon to produce an alkene



3) Reaction with metals

Haloalkane reacts with metals, to form a compound containing carbon - metal bond known as organometallic compounds.

a) Grignard reaction

When a solution of halo alkane in ether is treated with magnesium, we get alkyl magnesium halide known as Grignard reagent.

Example

$CH_3CH_2Br + Mg \xrightarrow{\text{Dry ether}} CH_3CH_2MgBr$

Ethyl bromide — Ethyl magnesium bromide

b) Reaction with active metals like sodium, lead etc

Haloalkane reacts with active metals like sodium, lead etc in the presence of dry ether to form organo metallic compounds.

$CH_3CH_2Br + 2Li \xrightarrow{\text{Dry ether}} CH_3CH_2Li + LiBr$

Ethyl bromide — Ethyl Lithium

$4CH_3CH_2Br + 4Na/Pb \longrightarrow (CH_3CH_2)_4Pb + 4NaBr + 3Pb$

Ethyl bromide — Sodium - lead alloy — Tetra ethyl lead (TEL)

4) Reduction reactions

Haloalkanes are reduced to alkanes by treating with H_2 in the presence of metal catalyst like nickel, palladium etc or with hydroiodic acid in the presence of red phosphorous.

$CH_3CH_2Br + H_2 \xrightarrow[523\ K]{Ni(or)Pd} CH_3\text{-}CH_3 + HBr$

Bromo ethane — Ethane

$CH_3CH_2I + HI \xrightarrow{Red\ P} CH_3\text{-}CH_3 + I_2$

Iodo ethane — Ethane

The chemistry of haloalkane can be well understood by the following flowchart.

14.3.6 Uses of haloalkane

Chloroform: As a solvent in pharmaceutical industry and for producing pesticides and drugs As an anaesthetic.

As a preservative for anatomical specimens.

Iodoform: Iodoform is used as an antiseptic for dressing wounds.